

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA1610
COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

<i>Criteria</i>	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

*I TEJASWINI.M 192411079 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 2: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

Answer

K-Means and **Hierarchical Clustering** are two commonly used clustering techniques.

Feature	K-Means	Hierarchical
Approach	Partition-based	Tree-based
Number of clusters	Must be specified	Can be decided using dendrogram
Speed	Faster	Slower
Scalability	Good for large datasets	Less suitable for very large datasets
Output	Fixed clusters	Dendrogram
Computation	Relatively low	Relatively high

For this dataset, K-Means can group customers according to their **income and spending score**.

For example:

- C1 and C4 have lower income and relatively high spending scores.
- C2 and C5 have high income and lower spending scores.
- C3 has high spending and medium-high income.

Performance

K-Means is generally faster because it repeatedly assigns points to cluster centroids.

Hierarchical clustering requires calculating distances between observations and progressively merging or splitting clusters, making it more computationally expensive.

Scalability

For a small dataset like this, **both methods are suitable**. However, for a large customer database, **K-Means is generally more scalable**.

Conclusion

For this dataset, both algorithms can provide useful customer segmentation. However, **K-Means is preferred for large-scale customer data because of its speed and scalability**, while hierarchical clustering is useful when a hierarchy of customer groups is required.

Q2. Dataset: Document Data

Document Word1 Word2 Word3

D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)

Answer

K-Means assigns each document to **exactly one cluster**.

For example:

D1 → Cluster 1

This is called **hard clustering**.

EM can provide a **probability of membership** for each document.

For example:

Document Cluster 1 Cluster 2

D1	0.80	0.20
D2	0.90	0.10
D3	0.10	0.90
D4	0.05	0.95
D5	0.75	0.25

These probabilities are illustrative; the question does not provide calculated EM probabilities.

Comparison

Feature	K-Means	EM
Cluster assignment	Hard	Soft
Overlapping clusters	Limited	Better

Feature	K-Means	EM
Output	Cluster label	Membership probability
Complexity	Lower	Higher
Flexibility	Lower	Higher

Documents D1, D2 and D5 have similar word distributions, while D3 and D4 have a different distribution.

Conclusion

For clearly separated documents, K-Means is simple and effective. However, when documents may belong partially to multiple topics, **EM is more suitable because it provides probability-based cluster membership.**

Q3. Dataset: Geographic Data

Point Latitude Longitude

P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Answer

Distance metrics determine how similarity or dissimilarity between data points is measured.

Euclidean Distance

The Euclidean distance between two points is:

It represents the **straight-line distance** between two points.

Manhattan Distance

Manhattan distance is:

It measures distance along individual coordinate directions.

Comparison

Feature	Euclidean	Manhattan
Distance	Straight-line	Grid/path-based
Formula	Square root of squared differences	Sum of absolute differences
Sensitivity	More affected by large differences	Less affected by individual large differences
Suitable for	Continuous numerical data	Grid-like/high-dimensional data

For example, between P1 and P2:

Manhattan distance:

Thus, the numerical distance depends on the selected metric.

Impact on Clustering

Changing the distance metric can change which points are considered similar.

- Euclidean distance tends to form clusters based on straight-line proximity.

- Manhattan distance can produce different cluster boundaries because it sums coordinate-wise differences.

Conclusion

Both metrics can be used for this geographic dataset. The choice of distance metric can affect cluster membership and therefore the final clustering result.

Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)

Answer

Hierarchical clustering has two major approaches:

Agglomerative Clustering

Agglomerative clustering follows a **bottom-up approach**.

Initially:

Every product = one cluster

Then similar clusters are progressively merged.

For this dataset:

- P1 and P3 have similar high sales and frequency.
- P2 and P5 have similar sales and frequency.
- P4 has lower sales and frequency.

Therefore, agglomerative clustering can naturally identify groups of similar products.

Divisive Clustering

Divisive clustering follows a **top-down approach**.

Initially:

All products = one cluster

The algorithm then repeatedly divides the cluster into smaller groups.

Comparison

Feature	Agglomerative	Divisive
Approach	Bottom-up	Top-down
Starting point	Individual points	One large cluster
Operation	Merge	Split
Complexity	Generally simpler	Generally more computationally expensive
Use	Commonly used	Useful when major divisions are important

Suitability

For this small retail dataset, **both methods can be applied**.

Agglomerative clustering is particularly suitable because the dataset contains only five products and similar products can easily be merged based on sales and purchase frequency.

Conclusion

Agglomerative clustering is a suitable choice for this dataset because it can progressively combine similar products and reveal natural product groups. Divisive clustering is also possible but may be more computationally demanding.

Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)

Answer

Centroid-Based Clustering

Centroid-based methods such as **K-Means** group data points according to their distance from cluster centroids.

For this dataset:

- I1, I3 and I5 have similar high prices and ratings.
- I2 and I4 have lower prices and ratings.

Therefore, numerical attributes can help form clusters.

Density-Based Clustering

Density-based algorithms such as **DBSCAN** identify areas containing many closely located points.

They can also identify:

- Dense regions
- Sparse regions
- Outliers

Comparison

Feature	Centroid-Based	Density-Based
Example	K-Means	DBSCAN
Main idea	Distance from centroid	Density of points
Cluster shape	Usually compact/spherical	Can identify irregular shapes
Outlier handling	Poorer	Better
Number of clusters	Usually specified	Can determine based on density
Mixed data	Requires suitable numerical encoding	Requires suitable distance representation

Mixed Data Problem

This dataset contains:

- **Price** → Numerical
- **Rating** → Numerical
- **Category** → Categorical

Standard K-Means cannot directly process the categorical **Category** attribute without converting it into a suitable numerical representation.

Therefore, preprocessing is required.

For example, Category A and B can be encoded appropriately before clustering.

Effectiveness

For this dataset, centroid-based clustering can work well after proper preprocessing because the numerical values show clear groups.

Density-based clustering can be useful when the data contains irregular clusters or outliers. However, its effectiveness also depends on choosing an appropriate distance measure and representing the categorical variable correctly.

Conclusion

Centroid-based clustering is simple and effective for compact numerical groups, while density-based clustering is better for irregularly shaped clusters and outlier detection. Since this dataset contains both numerical and categorical attributes, **proper preprocessing and an appropriate distance measure are essential before applying either technique.**